Artificial Intelligence 2: Introduction

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Outline of Topics

Lectures times/venues

- Lecturers
  - Shan He: Weeks 1-5 (Email: s.he@cs.bham.ac.uk, Office hour: Thursday 14:00 - 16:00)
  - Hyung Jin Chang Weeks 6-7
  - Miqing: Weeks 8-10
- Lecture structure
  - Two lectures per week: Friday morning 10-12
  - Two tutorials (TAs): Wednesday afternoon from week 3
    - Help-desk: exercise questions
    - Programming tutorial: AI programming tutorial

What is AI?

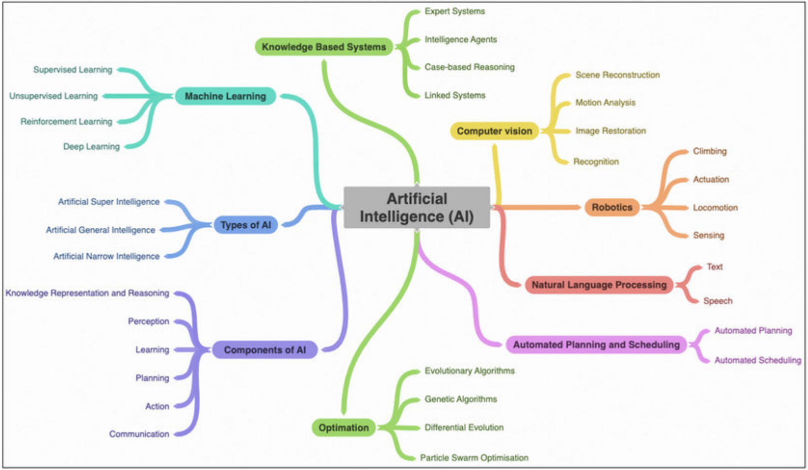

Figure 1: M. Regona et.al, Opportunities and Adoption Challenges of AI in the Construction Industry: A PRISMA Review , JOITMC. 8(1):45, 2022

Aims of the module

We will learn:

- Probabilistic machine learning (Weeks 1-5):
    - Maximum likelihood – Logistic Regression and Classification
    - Information theory: entropy, cross entropy, mutual information, information and decision tree learning and feature selection.
    - Probabilistic Graphical Models: Bayesian networks
- Computer Vision (Weeks 6-7)
- Automated planning and scheduling (Weeks 8-10): Advanced search algorithms and Constraint Satisfaction Problems
- Revision (Week 11)

Learning outcomes

By the end of the module, we should be able to:

- Conceptually understand frameworks for consistent treatment of uncertainty in AI
- Understand principles of inference and fitting to data in AI models under uncertainty
- Describe and understand randomised search and optimisation techniques utilised in AI

To better prepare you for the new era of AI.

Exam and Continuous Assessment

Main Assessments: 1.5 hour examination (80%) and continuous assessment (20%)

Continuous assessment: two online (Canvas) quizzes, 10% each.

- Probabilistic Machine Learning (Week 5)
- Advanced Search and CSP and (Week 10)

Supplementary Assessments: 1.5 hour examination

Text books

Probability and statistics

- Probability Theory: The Logic of Science

Information theory

- Information Theory, Inference, and Learning Algorithms

AI and ML:

- Artificial Intelligence: Foundations of Computational Agents, second edition, Cambridge University Press 2017
- Mathematics for Machine Learning
- Bayesian Reasoning and Machine Learning

Let's chat about ChatGPT (Large Language Models) and AI

Discussion: What do you know about about LLMs such as ChatGPT?

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The mission of OpenAI

"*Our mission is to ensure that artificial general intelligence – AI systems that are generally smarter than humans – benefits all of humanity.*"

What is Artificial General Intelligence

No broadly accepted definition, but here is a list of what AGI is required:

- Reason, use strategy, solve puzzles, and make judgments under uncertainty
- Represent knowledge, including common sense knowledge
- Plan
- Learn
- Communicate in natural language
- Integrate these skills in completion of any given goal

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Discussion:

- What have current LLMs achieved in terms the AGI requirements?
- Do you think we can achieve Artificial General Intelligence?
  - If not, why?
  - If yes, justify why we should achieve it.

Problems with ChatGPT

Read:

- Testing the cognitive limits of large language models
- ChatGPT failure archive

What's your opinion about AGI after reading?

More Discussions

- How AI will impact our society, e.g., jobs, but most importantly, your life after 1 years and 5 years?
- How shall we prepare for the impact?

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- OpenAI CEO Sam Altman's 2024 Jan. interview at Davos.
- The summary of the interview from CNBC News.